

title

Observación 1. Sea $f \in \mathcal{H}(\Omega)$, entonces $\bar{f}$ es antiholomorfa en Ω y, además, $\forall z \in \Omega$:

$$\begin{aligned}
 \frac{\partial \bar{f}}{\partial \bar{z}}(z) &= \frac{1}{2} \left(\frac{\partial \bar{f}}{\partial x}(z) + i \frac{\partial \bar{f}}{\partial y}(z) \right) = \frac{1}{2} \left(\frac{\partial u}{\partial x}(z) - i \frac{\partial v}{\partial x}(z) + i \frac{\partial u}{\partial y}(z) + \frac{\partial v}{\partial y}(z) \right) \\
 &\stackrel{\text{C-R}}{=} \frac{1}{2} \left(\frac{\partial u}{\partial x}(z) - i \frac{\partial v}{\partial x}(z) + i \left(-\frac{\partial v}{\partial x}(z) \right) + \frac{\partial u}{\partial x}(z) \right) = \frac{1}{2} \left(2 \frac{\partial u}{\partial x}(z) - 2i \frac{\partial v}{\partial x}(z) \right) \\
 &= \frac{\partial u}{\partial x}(z) - i \frac{\partial v}{\partial x}(z) = \overline{\frac{\partial f}{\partial z}(z)} = \overline{f'(z)}.
 \end{aligned}$$